

140. human TRIM31 ZN_FING "B box-type" (42 aa) and mouse Q99PN3

TOOLS: kmerseek hp_thomas_dill_no_c2 k19 (chosen for ZN_FING in Stage 0) and hp_thomas_dill2 k19; phmmer, MMseqs2, MMseqs2 iterative, Foldseek, ProstT5, Reseek; Kyte-Doolittle scan

- kmerseek: call lands on the feature (>= 80% of call inside, >= 30% covered)

call mostly inside the feature, lower IoU than kmerseek

call overlaps the feature but < 50% of the call is inside it

call reaches kmerseek's IoU or higher

no call of this feature type overlaps the feature

the tool's best call is on another target protein

Kyte-Doolittle scan: no target protein, it searches nothing

the Swiss-Prot feature being scored (human) / transferred from (target)

other Swiss-Prot features, labelled by type
- human TRIM31 (Q9BZY9)

kmerseek hp_thomas_dill_no_c2 k19, chosen for ZN_FING (IoU 0.64)

kmerseek hp_thomas_dill2 k19, best other alphabet (IoU 0.64)

phmmer (IoU 0.64)

MMseqs2 (IoU 0.51)

MMseqs2 iterative (IoU 0.82)

Foldseek (IoU 0.35)

ProstT5 (IoU 0.32)

Reseek (IoU 0.49)

Kyte-Doolittle scan

residue position on human TRIM31 (Q9BZY9) (aa; protein is 425 aa, window shown)

Legend: COILED, ZN_FING, no call
- mouse Q99PN3

kmerseek hp_thomas_dill_no_c2 k19, chosen for ZN_FING

kmerseek hp_thomas_dill2 k19, best other alphabet

phmmer

MMseqs2

MMseqs2 iterative

Foldseek

ProstT5

Reseek

Kyte-Doolittle scan

residue position on mouse Q99PN3 (aa; protein is 545 aa, window shown)

Legend: its best call is on another target protein, no target: the scan reads the human sequence only
- Hypothesis: kmerseek places the human ZN_FING feature "B box-type" on its mouse counterpart and the structure tools do not.
Conclusion: kmerseek IoU 0.64, 16 of 27 residues identical; Foldseek spills; ProstT5 spills; Reseek spills; phmmer inside, lower IoU; MMseqs2 inside, lower IoU; MMseqs2 iterative equal or higher IoU; Kyte-Doolittle scan did not land; pLDDT 88, disorder fraction 0.0.